MAT092: Remedial Course in Mathematics

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LECTURE-03

Set Theory Review: Basic Level

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(b) $\{2\} \subseteq A$
— Yes, every element of $\{2\}$ (only 2) belongs to A.
(c) $\{2\} \in \mathscr{P}(A)$
— Yes, because $\{2\}$ is a subset of A, so it appears in the power set.

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Let $B = \{1, 3, 5, 7\}$, $C = \{3, 4, 5, 6\}$.

$$A \cup B = ?$$
$$A \cap B = ?$$
$$A - B = ?$$

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$$A \cup B = \{1, 3, 4, 5, 6, 7\}$$
$$A \cap B = \{3, 5\}$$
$$A - B = \{1, 7\}$$

Example

If $n(A) = 20$, $n(B) = 15$, $n(A \cup B) = 25$, find $n(A \cap B)$.

Example

If $n(A) = 20,\ n(B) = 15,\ n(A \cup B) = 25$, find $n(A \cap B)$.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) \Rightarrow 25 = 20 + 15 - n(A \cap B)$$

$$\Rightarrow n(A \cap B) = 10$$

Example

Write the power set $\mathscr{P}(\{a, b\})$.

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$$\mathscr{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

(Every subset is listed, from empty to full set.)

Example

True or False:

$\emptyset \in \{\emptyset\}$ - ?

$\emptyset \subseteq \{\emptyset\}$ - ?

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True or False:

$\emptyset \in \{\emptyset\}$ — True, because the empty set is an element of the set $\{\emptyset\}$.

$\emptyset \subseteq \{\emptyset\}$ — True, the empty set is a subset of every set.

Example

If $U = \{1,2,3,4,5,6\}$, $A = \{1,2,3\}$, find A'.

$$A' = U - A = \{4,5,6\}$$

Find $\mathscr{P}(\emptyset)$.

$$\mathscr{P}(\emptyset) = \{\emptyset\} \quad \text{(only one subset — itself)}.$$

Set Theory Review: Moderate Level

$U = \{1,2,3,4,5,6,7,8\}$, $A = \{1,3,5,7\}$, $B = \{3,4,5,6\}$.

$$A' = U - A = \{2,4,6,8\}, \quad A \triangle B = (A - B) \cup (B - A) = \{1,4,6,7\}.$$

(Symmetric difference = elements not common to both.)

Example

If $A = \{x, y, z\}$, list all subsets.

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If $A = \{x, y, z\}$, list all subsets.

$$\mathscr{P}(A) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}$$

There are $2^{|A|} = 2^3 = 8$ subsets.

Example

$A = \{1, 2\},\ B = \{x, y, z\}.$

Example

$A = \{1, 2\},\ B = \{x, y, z\}.$

$A \times B = \{(1,x),(1,y),(1,z),(2,x),(2,y),(2,z)\}, \quad B \times A = \{(x,1),(x,2),(y,1)$

Not equal because order of elements in pairs matters.